

Image Forgery Detection

Project Proposal — EDS6364 24851 MAIN - Digital Image Processing

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1. Project Title

Image Forgery Detection Using Classical Image Processing Techniques

2. Image / Dataset Information

Dataset: CASIA v2.0 Image Tampering Detection Dataset.

Application Domain: Digital forensics / photography — detecting manipulated (spliced and copy-move forged) photographs, with relevance to misinformation detection, insurance-fraud review, and scientific-image integrity checks.

Source: [Kaggle — CASIA 2.0 Image Tampering Detection Dataset](#), originally released by the Institute of Automation, Chinese Academy of Sciences for academic/research use. It is a widely cited benchmark, so published work exists to compare our results against.

Description: 12,614 JPEG images — 7,491 authentic and 5,123 tampered — covering two forgery types (splicing and copy-move) at varying resolutions. A companion ground-truth mask set (also on Kaggle) marks the exact tampered pixels for a subset of images, which we plan to use to score localization accuracy (e.g., Intersection-over-Union) rather than only whole-image classification.

3. Project Objective

The goal is to determine whether a given image has been digitally tampered with, and to localize the manipulated region, using classical, interpretable image-processing signals rather than a black-box deep model. The system will (a) classify each image as authentic or forged, and (b) output a visual map highlighting the suspected tampered region, evaluated against the dataset's ground-truth masks where available.

We selected this problem because it has direct real-world relevance, and because it produces clear, evaluable before-and-after results at every pipeline stage (original → ELA map → frequency/edge evidence → final localization), making it straightforward to demonstrate whether each technique actually helped. Since CASIA v2.0 covers two distinct forgery mechanisms, the project also lets us test whether one pipeline generalizes across both, or whether splicing and copy-move need different cues.

4. Planned Image Processing Pipeline

The pipeline below draws complementary evidence from all five technique categories; per-image features are fused and passed to a classical classifier, and the same intermediate maps are combined to localize the forged region.

Category	Technique & Tentative Parameters	Why it is useful here
Point Processing	Error Level Analysis (ELA): re-save the image at JPEG quality ≈ 90 , then take the per-pixel absolute intensity difference from the original, scaled $\times 10\text{--}15$ for visibility.	Tampered regions were pasted in after the original compression, so they re-compress at a different error rate than the untouched background. A pixel-wise intensity operation is the most direct way to expose that mismatch.
Histogram Processing	Histogram equalization / contrast stretching on the ELA map; chi-square comparison of local 32×32 -patch histograms across the image.	The raw ELA map is low-contrast and hard to read. Equalizing it makes faint artifacts visible; while comparing patch histograms flags regions whose intensity distribution statistically differs from their neighbors.
Spatial Filtering	3×3 Laplacian high-pass filter plus sliding 8×8 -window local variance, producing a per-patch noise-level map.	A spliced region usually comes from a different camera/sensor and carries a different noise signature. Spatial filtering converts that difference into a measurable, comparable quantity per patch.
Frequency-Domain Filtering	Block-wise 8×8 Discrete Cosine Transform (matching the JPEG block)	Re-saving a spliced region introduces a second, often misaligned compression grid. This

	grid); inspect AC-coefficient histograms for the double-quantization peaks that indicate double compression.	periodicity/discontinuity in the DCT coefficients is invisible in the spatial domain but shows up clearly in frequency space.
Edge & Corner Detection	Canny edge detection (thresholds $\approx 50/150$) around suspected boundaries; Harris corner response to find and match duplicated key points for copy-move regions.	Pasted regions often leave faint edge discontinuities at their border, and copy-move forgery literally duplicates a patch of the image — matching corner key points is a direct way to locate both copies.

Feature fusion & classification: ELA statistics, patch-histogram distances, local noise variance, DCT double-quantization scores, and edge/corner-match counts are concatenated into one feature vector per image and passed to an SVM (and, separately, a Random Forest) for authentic-vs-tampered classification; the same maps are thresholded and overlaid to produce the localization output.

Evaluation plan: classification accuracy, precision/recall, and confusion matrix on a held-out test split; localization quality via pixel-level IoU against ground-truth masks where available.

Implementation tools: Python with OpenCV (color conversion, filtering, edge/corner detection), NumPy (array/DCT operations), scikit-image (texture and additional segmentation utilities), scikit-learn (SVM / Random Forest, train-test evaluation), and Matplotlib for the before-and-after visualizations used to justify our technique and parameter choices.

5. Optional Deep Learning Extension

As a bonus extension, we may replace or augment the classical classifier with a CNN trained on the ELA-transformed images, using transfer learning with a pretrained backbone (e.g., ResNet18 or VGG16) fine-tuned on CASIA v2.0. This would let us directly compare classical feature-based classification against a deep-learning approach on the same pipeline outputs. This component is not required for the core project — the pipeline in Section 4 already satisfies the course requirements — but we may pursue it if time permits.

6. Anticipated Challenges & Mitigation

Parameter sensitivity: the ELA JPEG-quality level and the DCT/edge thresholds listed in Section 4 are starting points, not final values — small shifts in these can change detection accuracy noticeably, so we will sweep a short range of each on a held-out validation subset and report which value we settle on and why, as the course guidelines request.

Generalization across forgery types: splicing and copy-move forgeries leave different forensic traces (a splice changes compression/noise statistics, a copy-move duplicates content within the same image), so a single fused feature vector may favor one type. We plan to also report accuracy separately per forgery type to check whether the pipeline needs type-specific weighting.

False positives on visually busy images: photographs with complex textures or lighting can mimic tampering artifacts (e.g., natural noise or compression variation), which may inflate false-positive rates. We will inspect misclassified examples manually to see whether additional post-processing (e.g., smoothing the localization map) is needed.

7. Team Responsibilities

Smit Patel: dataset preparation and preprocessing, and implementation of the point-processing (ELA), histogram-processing, and spatial-filtering stages of the pipeline.

Ananay Gupta: implementation of the frequency-domain (DCT) and edge/corner-detection stages, feature fusion, and classifier training.

Both members will jointly handle evaluation, before-and-after visualizations, parameter justification, and the final report..